

Name: Kahathota Liyanage Yasindu Janiya Weerasinghe

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Course: Machine Learning/Artificial Intelligence

Predicting Appliance Energy Use in a Low-Energy House

1. Introduction

This project investigates the predictive modeling of appliance energy consumption using environmental and temporal data. The dataset utilized for this analysis originates from a study of a low-energy house in Belgium over a 4.5-month period (Candanedo et al., 2017). It contains 19,735 observations recorded at 10-minute intervals, comprising 29 features that primarily capture indoor and outdoor temperature and humidity, alongside weather station metrics such as atmospheric pressure and wind speed.

The target variable for this analysis is the *Appliances* column, representing the total energy used by appliances in Watt-hours (Wh) per 10-minute interval. The distribution of this variable is highly right-skewed; while the mean consumption sits at a low 97.7 Wh, the maximum surges to 1080 Wh. This extreme variance makes predictive modeling inherently challenging, as the target is dominated by low-energy baseline periods punctuated by sudden, massive energy spikes caused by random human behavior.

Given these characteristics, traditional linear regression is ill-equipped for this dataset. Appliance energy usage is not a steady, linear progression; it is driven by complex, non-linear triggers. For example, HVAC systems activate based on strict temperature thresholds, and cooking appliances are driven by specific occupancy routines tied to the time of day. Exploratory Data Analysis (EDA) confirmed this, revealing a severe right-skew in the target histogram and very low linear correlations between individual weather features and energy use. Consequently, this project leverages advanced non-linear and ensemble machine learning techniques to capture these threshold-driven relationships.

2. Methods

2.1 EDA Summary

Initial Exploratory Data Analysis revealed that the target variable is heavily right-skewed, with the vast majority of readings falling below 100 Wh. The correlation heatmap demonstrated very weak linear relationships between the target and individual independent variables. Furthermore, severe multicollinearity was observed among the independent variables, particularly between various indoor temperature and humidity sensors.

2.2 Preprocessing

The dataset was clean, with no missing values requiring imputation. Using the Interquartile Range (IQR) method, approximately 11% of the target data was flagged as statistical outliers. However, these outliers were intentionally kept in the dataset, as they represent legitimate, critical events (actual high-energy appliance usage) rather than sensor errors. To capture human routines, feature engineering was applied to extract `hour` and `day_of_week` from the raw timestamp. Finally, standard scaling (`StandardScaler`) was applied. The scaler was fit strictly on the training set and then used to transform the test set to prevent data leakage. Scaling was mathematically mandatory to ensure distance-based algorithms, such as Support Vector Regression (SVR), were not disproportionately skewed by features with large numerical magnitudes like atmospheric pressure.

2.3 Train/Test Split

The data was split into an 80% training set and a 20% testing set. Because the target variable contained extreme, rare energy spikes, a standard random split risked placing all the spikes in one set. To solve this, `pd.qcut` was used to bin the continuous target variable, allowing for a stratified split that ensured a proportionate distribution of extreme values in both sets. A strict 5-Fold Cross-Validation (`KFold` with `shuffle=True`) was utilized consistently across all model training phases to ensure robust, unbiased evaluation.

2.4 Models Selected

Four distinct regression models were evaluated to establish a performance hierarchy:

Linear Regression: Served as the baseline to prove that simple additive math is insufficient.

Decision Tree: Selected to capture non-linear, threshold-based logic (e.g., "if temperature < 20°C").

Support Vector Regression (SVR): Chosen for its ability to map features into high-dimensional space to capture non-linear geometric relationships.

Random Forest: Deployed to mitigate the high variance and overfitting of a single Decision Tree via ensemble averaging.

For hyperparameter tuning, `GridSearchCV` was used for the Decision Tree to exhaustively search a small parameter space. However, `RandomizedSearchCV` was deployed for the SVR and Random Forest models. This allowed the models to sample a broader range of hyperparameters within a strictly controlled computational budget, bypassing the massive processing time required for exhaustive grid searches on complex algorithms.

3. Results

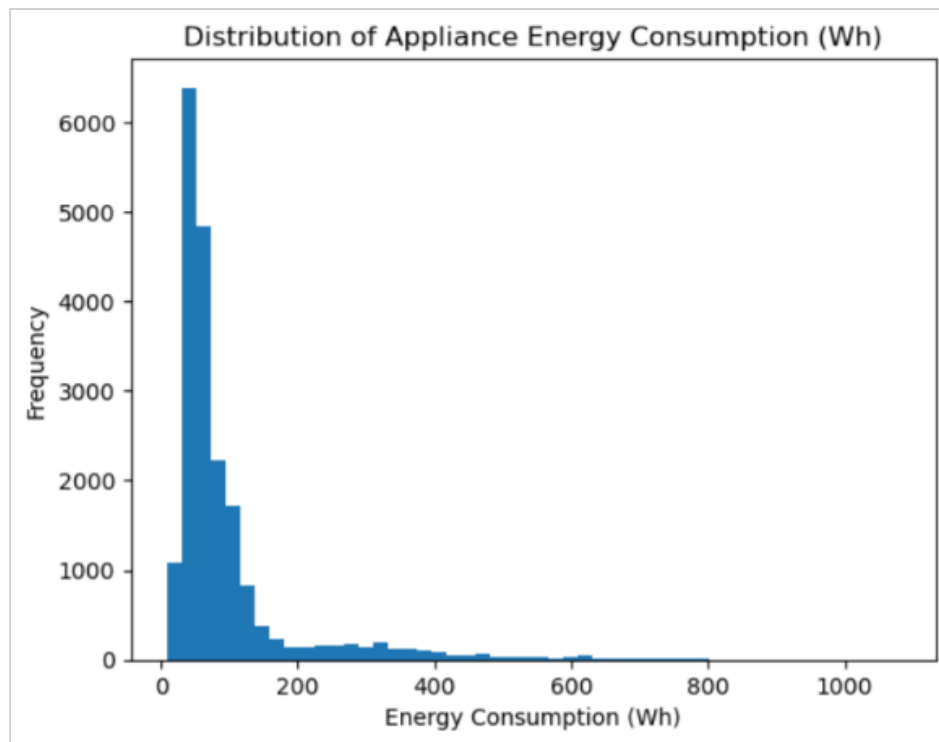
3.1 Model Comparison Table

Model	CV R ²	Train R ²	Test R ²	MAE (Wh)	RMSE (Wh)
Linear Regression	0.1624	0.1675	0.1696	53.16	93.17
Decision Tree	0.2721	0.5143	0.3336	41.30	83.47
SVR	0.2213	0.2618	0.2353	36.89	89.41
Random Forest	0.4800	0.9135	0.5305	33.07	70.05

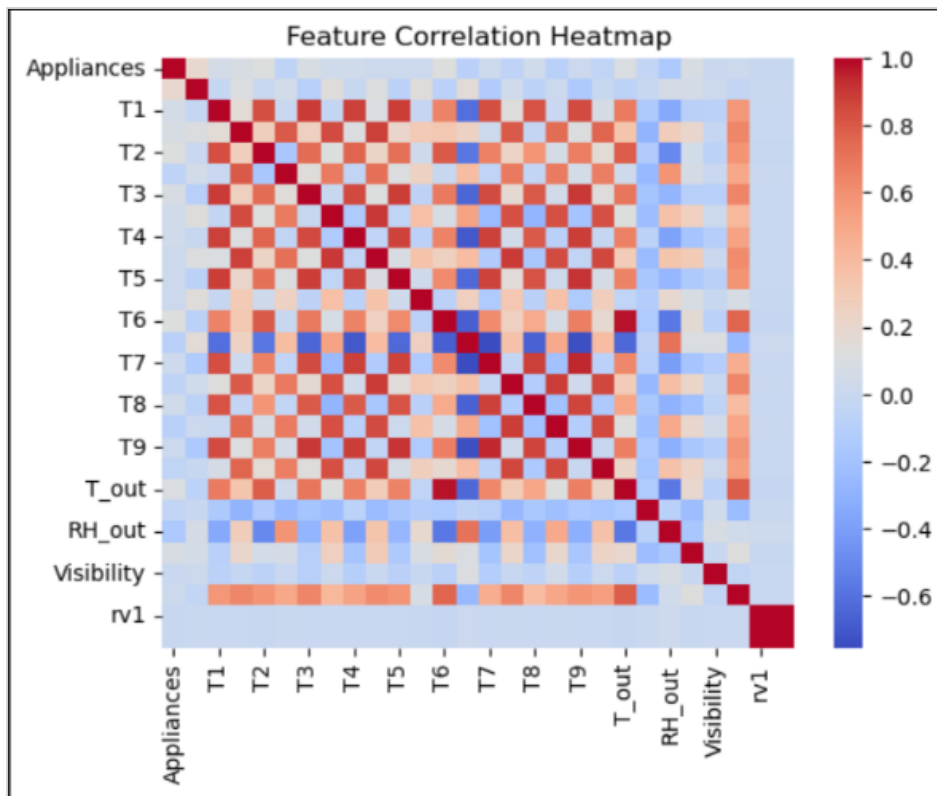
3.2 Best Model Performance

The Random Forest ensemble emerged as the undisputed optimal model across all metrics. It achieved a Test R² of 0.5305 and the lowest Mean Absolute Error of 33.07 Wh. This represents a more than threefold improvement in explained variance compared to the linear baseline (R² ~ 0.16). This massive leap in performance is attributed to the Bagging (Bootstrap Aggregating) technique. By averaging the predictions of 75 diverse decision trees, the ensemble successfully captured complex, non-linear human routines while smoothing out the rigid, overfitted thresholds of any single tree.

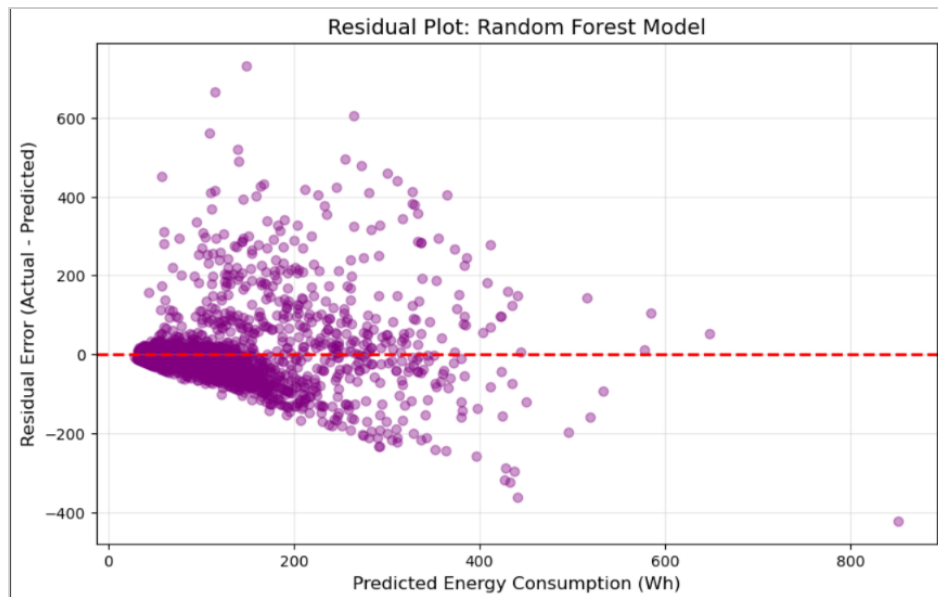
3.3 Key Plots



Target Distribution: This histogram illustrates the extreme right-skewed nature of the Appliances target variable. The dense clustering near zero represents the house's passive baseline energy, while the long tail visually demonstrates the sudden, extreme energy spikes that make predictive modeling difficult.



Feature Correlation Heatmap: This chart reveals that the engineered hour feature is the single most dominant predictor in the Random Forest model, confirming that daily human routines dictate energy consumption. Interestingly, day_of_week ranked dead last, proving that a 6:00 PM energy spike follows the same pattern regardless of whether it is a Tuesday or a Saturday.



Residual Analysis: The Random Forest residual plot displays a clear "fan-shape" (heteroscedasticity). While the model is highly accurate at predicting low-energy baseline periods (dense clustering around the zero-line), the residuals expand significantly as predicted energy increases, showing a systematic failure to accurately capture extreme peak loads.

4. Discussion

4.1 Error Analysis

An analysis of the top 10 largest prediction errors revealed a clear systematic vulnerability: the model heavily struggles with extreme energy spikes (values exceeding 500 Wh). In every major error instance, the Random Forest massively under-predicted the actual consumption. This is a mathematical consequence of the ensemble architecture; because the final prediction is an average of 75 trees, the model is inherently pulled toward the mean, naturally suppressing extreme maximums.

4.2 Limitations

This analysis is bound by several distinct limitations. First, the dataset relies on a single house over a short 4.5-month span, meaning the model has not learned full seasonal variations and cannot generalize to other climates. Second, the dataset lacks explicit behavioral features—the model is forced to guess human occupancy using time and lighting proxies, without knowing how many people are home or which specific appliances are running. Finally, using a standard randomized Train/Test split on time-series data fails to respect strict chronological order, technically allowing future data points to influence the prediction of past events.

4.3 Future Work

Future iterations of this predictive engine should implement a strict Time-Series split for validation to properly mimic real-world chronological forecasting. Algorithmically, upgrading to a Gradient Boosting model (such as XGBoost) could help correct the extreme-spike under-predictions by building trees sequentially to focus on previous errors, while an LSTM neural network could better capture sequential temporal dependencies. Ultimately, augmenting the dataset with direct IoT sensor data—such as explicit room occupancy counters and smart-plug appliance monitors—would provide the vital non-weather context required to fully close the prediction gap.

References

Candanedo, L. M., Feldmann, A., & Degemmis, D. (2017). Data driven prediction models of energy use of appliances in a low-energy house. *Energy and Buildings*, 145, 13–25. <https://doi.org/10.1016/j.enbuild.2017.03.040>

AI Usage Statement

I used Claude as a learning tool for conceptual explanations of machine learning techniques, hyperparameter tuning strategies, and guidance on project structure. All code execution, data analysis, visualizations, and written interpretations were produced and verified independently. No code was directly copied from AI-generated outputs without manual review and integration.